

SASTRA DEEMED UNIVERSITY

(A University under section 3 of the UGC Act, 1956)

End Semester Examinations

February 2022

Course Code: CIV101

Course: BASIC CIVIL ENGINEERING

Question Paper No. : UGF007

Max. Marks:100

PART -A

Answer all the questions

10 x 2 = 20 Marks

1. Contrast the importance of Water Resources Engineering and Management, Public Health and Environmental engineering branches of civil engineering fields.
2. A plot of ground 380 m^2 , has a residential building with built up area of 160 m^2 in ground floor and 130 m^2 in first floor. Compute the Floor Space Index.
3. Identify the term - 43 Grade cement and M15 concrete.
4. Outline the differences of a load bearing structure and a framed structure.
5. Under what circumstances, the queen closers and king closers are used in the brick masonry construction works?
6. Bring out the purposes, width of the carriage way and shoulders in a two lane bitumen surfaced roads.
7. Convert the following Whole Circle Bearings (WCB) to Reduced Bearings (RB)
(a) $160^\circ 30'$ (b) $232^\circ 30'$ (c) 90° (d) 270°

8. State the various applications of Remote Sensing.
9. List the components of Gravity dam.
10. Specify the necessity of watershed management.

PART- B

Answer all the questions

4 x 15 = 60 Marks

11. (a) Briefly describe the various functions of Civil Engineers, which has been performed before, during and after construction of a residential project. (10)
- (b) Outline the importance of the planning principles grouping and roominess, which ensures the comfort of living in a well planned building. (5)

(OR)

12. (a) Describe the various types of cement, practiced in civil construction projects with their applications. (10)
 - (b) Identify the commercial forms of steel, which are available in market? (5)
13. (a) Explain the various types of shallow foundation with neat sketches. (10)
 - (b) Classify the reinforced cement concrete beams according to the support conditions. (5)

(OR)

14. (a) With neat sketches, briefly describe the English and Flemish bonds adopted in brick masonry constructions. (10)
 - (b) Sketch a panelled door and name its components. (5)
15. The following perpendicular offsets were taken at 8m intervals from a survey line to an irregular boundary line.

3.50m, 4.60m, 8.30m, 5.80m, 6.90m, 7.80m, 4.30m, 5.80m, 4.40m

Calculate the area enclosed between the survey line, the irregular boundary line, first offset and the last offsets by the application of Average ordinate rule, Trapezoidal rule and Simpson's rule.

(OR)

16. The following staff readings were taken with a level, the instrument having been moved after third and seventh readings.

2.400, 1.730, 0.650, 2.900, 2.200, 1.420, 0.800, 0.510, 1.490, 0.990

Enter the above readings in a page of level book and calculate the reduced levels of points using Height of collimation method & Rise and fall method, if the first staff reading was taken with a staff held on a bench mark of 200.00m.

17. (a) Briefly describe the various types of structures involved in the watershed management works. (10)

(b) Outline any one method of rain water harvesting carried out in a residential colony area. (5)

(OR)

18. (a) With neat sketches, Explain the various types of Earthen Dams. (10)

(b) State the necessity and the important benefits of irrigation. (5)

PART -C

Answer the following

1 x 20 = 20 Marks

19. (a) Contrast the important points to be considered in the site selection process of a commercial building project. (10)

(b) With neat sketches, describe a dog legged stair and a straight flight stair. (10)

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